John Hartquist

Software Engineer · AI/ML Systems

Pleasant Hill, CA | john@hartquist.com | (805) 305-4168 | johnhartquist.com | github.com/jhartquist

Summary

Software engineer building systems from prototype to production. 14 years shipping software, primarily at startups, with the last 8+ focused on ML and AI. Previously CTO at TakeOne, owning an ML video pipeline end to end across mobile, web, and cloud. Open-source and independent work includes training real-time guitar transcription models, porting LLM and audio inference to Rust, and hacking on AI developer tools. Adept at steering coding agents toward robust, verifiable, and maintainable systems.

Experience

Staff Founding Engineer, Foam – San Francisco, CA — Aug 2026 – Oct 2026

- Built Ruby and Node.js OpenTelemetry auto-instrumentation SDKs powering Foam's AI agents: fail-safe APIs, coexistence with APM vendors (Datadog, Sentry, New Relic), PII-redacted network and database capture, and 100% branch coverage.
- Built cross-language (TypeScript, Python, Ruby) end-to-end verification harness, translating requirements into 1,700+ audited tests across relevant library and runtime versions; surfaced upstream bugs and patched cross-vendor conflicts.

Senior SWE (2023) → Staff SWE (2024) → CTO (2025), TakeOne – Remote — Aug 2023 – Mar 2026

- Shipped features across TakeOne's ML video pipeline, processing 10,000+ phone-recorded takes from 1,500+ users: video matting and segmentation (SAM, Track-Anything, GroundingDINO), skin-tone correction, audio cleanup (DeepFilterNet + FFmpeg loudness normalization), and transcription (Whisper, Deepgram).
- Built LLM video-editing pipeline with validation gates, LLM-as-judge, and retry-with-feedback.
- Built the full-stack customer portal solo (TypeScript, Next.js), where client admins review and publish videos.
- Built internal tools in Rust for bulk user onboarding, provisioning, data validation, and API testing.
- Wound the company down in March 2026.

Senior Software Engineer, Lowe's Innovation Labs – Kirkland, WA — Sept 2021 – July 2023

- Prototyped AR measurement and virtual-object simulation in Unity for in-store and at-home use cases.
- Built React / Babylon.js inspector and ingestion tool for 3D home assets; published a custom open-source product viewer.
- Built a room-scanning prototype using Apple's RoomPlan API to capture 3D room geometry.

Senior Software Engineer — ML Platform, Sigma IQ – Seattle, WA — Apr 2019 – June 2021

- Productionized a financial-reconciliation ML model (Python, XGBoost) and migrated the pipeline to Spark (Scala), achieving a 100x+ speedup over the prior Python implementation.
- Built dataset, training, and evaluation tooling to iterate on reconciliation model accuracy.
- Built a full-stack platform for running Jupyter notebooks in production (TypeScript, React, Next.js, Docker, GCP).

Independent, Self-Directed ML Deep Dive – San Francisco, CA — Jan 2018 – Apr 2019

- Studied classic ML, deep learning, and reinforcement learning full-time for a year.
- Built `fastai_audio` (194★), a library for on-the-fly GPU spectrogram generation during training; write-up.
- Placed top 8% (39th of 556 teams) in Kaggle's Freesound audio-tagging challenge with CNN ensembles.

Senior Software Engineer, Ruvixx, Inc. – San Francisco, CA — Apr 2015 – Dec 2017

- Built licensing and brand-protection platform (Ruby on Rails, Angular) used by HDMI, Dolby, and Philips.

Software Engineer, Emjoyment – Mountain View, CA May 2014 – Apr 2015

- Built iOS app (Objective-C) and Python/Django backend services for an early-stage 'Tinder for Jobs' startup.

Software Development Engineer, Amazon.com – San Luis Obispo, CA July 2012 – May 2014

- Built internal services for the CreateSpace Royalties & Financials team (Java, Spring, AWS) using test-driven development.

Selected Independent Research & Open Source

system2 — idiomatic Rust SDK for TypeSafe's Jev – github.com/jhartquist/system2

- Composes System 1 judgments into System 2 decisions in plain Rust, batching many questions per request.
- Answers come back as the caller's own types, from derived enums to arbitrary structs, with calibrated probabilities intact.
- Procedural macros validate literal questions at compile time; questions from data and every answer are checked at runtime.

resonators — real-time spectral analysis in Rust (97★) – github.com/jhartquist/resonators

- Rust implementation of Alexandre François's Resonate algorithm: per-bin spectral analysis without FFTs or windowing.
- Cross-platform: native on macOS, Linux, iOS, plus browser (WASM). Python (PyO3) and JavaScript bindings.
- Outperforms the Apple-only C++ reference implementation by 1.6x on M2 Max.
- Live WebGL streaming-spectrogram demo and write-up at johnhartquist.com/resonators.

claude-remotion-kickstart (122★) – github.com/jhartquist/claude-remotion-kickstart

- Claude Code + Remotion starter kit for programmatic video; walkthrough video has 43,000+ YouTube views.

groxide — Rust docs CLI for AI coding agents – github.com/jhartquist/groxide

- Converts rustdoc JSON into markdown so AI coding agents can query crate docs without spending tokens parsing HTML.

rloop — DAG-based task executor for Claude Code – github.com/jhartquist/rloop

- Runs tasks autonomously in isolated git worktrees with verification (tests, clippy, build) gating merges.
- Retry-with-learnings: failed attempts produce structured findings that feed forward into the next attempt's context.

candle-pitch — pitch detection models ported to Rust – github.com/jhartquist/candle-pitch

- End-to-end Rust ports of CREPE, Swift-F0, and Spotify's Basic Pitch via Hugging Face's Candle framework.
- Numerical-parity tested against the original TensorFlow / PyTorch implementations.

Real-Time Musical Analysis (RTMA)

- Cal Poly MS thesis (2012): real-time polyphonic guitar transcription using Non-Negative Matrix Factorization.
- Full Rust rewrite, GuitarSet evaluation, neural experiments with HCQT features and Mamba-family architectures.
- Adapted Spotify's Basic Pitch to run in real-time in the browser via `resonators` front-end.

Education

California Polytechnic State University, BS + MS in Computer Science – San Luis Obispo, CA Sept 2007 – June 2012

- Thesis: *Real-time Musical Analysis of Polyphonic Guitar Audio*

Skills

Languages: Python, TypeScript, Rust, Ruby; previously Swift, Kotlin, Scala, Java, C#

ML: PyTorch, Candle, Burn, TensorFlow, FastAI, Weights & Biases, XGBoost, ONNX, NumPy, pandas, Polars, Spark

AI: Claude API, OpenAI API, Gemini, Amazon Bedrock, TypeSafe Jev, MCP, Claude Code, Codex, Pydantic

Web: Next.js, React, Node.js, Three.js, React Three Fiber, Rust + WASM, FastAPI, Django, Axum, Tauri, Dioxus

Infrastructure: AWS, GCP, Lambda Cloud, Vercel, Supabase, Postgres, Docker, OpenTelemetry

Media: FFmpeg, Remotion, OpenCV, Segment Anything, Veo 3, Whisper, Deepgram, DeepFilterNet, Web Audio